

Finite-Dimensional Extensions of Nonseparable Compact-Operator Algebras Fail BCP

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June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

The continuous-trace positive packet proved the expected criterion under a separable-fibre hypothesis. The following elementary subcase shows that this hypothesis is necessary, even for a one-point primitive spectrum. The same argument also survives arbitrary finite-dimensional extensions of the form $\mathcal{K}(H) + F$.

Theorem 1. *Let H be a nonzero Hilbert space, let $F \subseteq \mathcal{B}(H)$ be a finite-dimensional C^* -subalgebra, and put*

$$A_F = \mathcal{K}(H) + F.$$

Then A_F has BCP if and only if H is separable. Equivalently, A_F has UBCP if and only if H is separable.

Corollary 1. *There is a type-I continuous-trace C^* -algebra with one-point primitive ideal space and without BCP: take $A = \mathcal{K}(H)$ for any nonseparable Hilbert space H . Thus countability of $\text{Prim}(A)$, or even the one-point continuous-trace case, does not imply BCP unless the fibre is separable.*

More generally, every finite-dimensional extension $\mathcal{K}(H) + F$ with nonseparable H fails BCP. Taking $F = \mathbb{C}1$ gives a unital type-I C^ -algebra with finite primitive ideal space and without BCP.*

Proof. The algebra $\mathcal{K}(H)$ is elementary, hence type I and continuous trace. It is simple for every nonzero Hilbert space H , so $\text{Prim}(\mathcal{K}(H))$ is a singleton. Each $\mathcal{K}(H) + F$ is an extension of a finite-dimensional C^* -algebra by $\mathcal{K}(H)$, hence is type I and has finite primitive ideal space. The failure of BCP in the nonseparable case is Theorem 1. $\square$

Idea of the Proof

The positive direction is just separability: if H is separable, then $\mathcal{K}(H)$ is a separable Banach space, hence has UBCP.

For the negative direction, fix countably many proposed centers $(T_n) \subset \mathcal{K}(H)$. Each compact operator has separable range, and so does its adjoint. The closed span of all ranges of T_n and T_n^* is therefore separable. In a nonseparable Hilbert space choose a unit vector ξ orthogonal to that span. The rank-one projection p_ξ is orthogonal to every T_n on both sides:

$$p_\xi T_n = T_n p_\xi = 0.$$

Thus $p_\xi - T_n$ is block diagonal with blocks 1 and $-T_n$, so

$$\|p_\xi - T_n\| = \max(1, \|T_n\|) \geq \|T_n\|.$$

No countable family of balls $B(T_n, \|T_n\|)$ can cover the unit sphere.

For a finite-dimensional extension $\mathcal{K}(H) + F$, decompose the finite-dimensional representation of F into matrix blocks with multiplicity spaces. Since H is nonseparable, one multiplicity space is nonseparable. Choose a vector η in that multiplicity space orthogonal to the coordinate supports of the compact parts of the centers. The finite-dimensional slice over η reduces all compact parts and carries the same F -matrix block as infinitely many nearby slices in the complement. Thus the norm of each center is already visible on the complement of the chosen slice, while a rank-one projection inside the chosen slice escapes the corresponding BCP ball.

Proof

We use the standard equivalent form of BCP: a normed space X has BCP iff there are countably many centers $x_n \in X$ such that

$$S_X \subseteq \bigcup_{n=1}^{\infty} B(x_n, \|x_n\|).$$

Indeed, enlarging any open ball not containing the origin to radius equal to the norm of its center still gives an open ball missing the origin.

Lemma 1. *If $T \in \mathcal{K}(H)$, then $\overline{T(H)}$ is separable.*

Proof. The image $T(B_H)$ of the unit ball is relatively compact. Its closure is a compact metric space, hence separable. Since

$$T(H) = \bigcup_{m=1}^{\infty} m T(B_H),$$

the closed linear span $\overline{T(H)}$ is separable. $\square$

Lemma 2. *Let E be a nonzero finite-dimensional Hilbert space and let K be an infinite-dimensional Hilbert space. Let Y be a Hilbert space containing $E \otimes K$ isometrically. If $S \in \mathcal{B}(E)$ and*

$$R = (S \otimes 1_K) + L : E \otimes K \rightarrow Y,$$

where $L : E \otimes K \rightarrow Y$ is compact, then

$$\|R\| \geq \|S\|.$$

Proof. Choose a unit vector $v \in E$ with $\|Sv\| = \|S\|$, and choose an orthonormal sequence (η_m) in K . The sequence $(v \otimes \eta_m)$ is weakly null, and compact operators send weakly null sequences to norm-null sequences. Hence

$$\|R(v \otimes \eta_m)\| \geq \|Sv \otimes \eta_m\| - \|L(v \otimes \eta_m)\| \rightarrow \|S\|,$$

which gives the claimed lower bound. $\square$

Proof of Theorem 1. If H is separable, then $\mathcal{K}(H)$ is separable: finite-rank operators with rational complex matrix entries relative to a fixed countable orthonormal basis are norm dense in $\mathcal{K}(H)$. Since F is finite-dimensional, $A_F = \mathcal{K}(H) + F$ is separable. Every separable normed space has UBCP: choose a countable dense set (d_j) in the unit sphere and use centers $2d_j$ with radius $3/2$. Hence A_F has UBCP, and therefore BCP.

Assume now that H is nonseparable, and let (a_n) be any countable family of centers in A_F . Write

$$a_n = f_n + T_n, \quad f_n \in F, \quad T_n \in \mathcal{K}(H).$$

By finite-dimensional representation theory, after a unitary identification we may write

$$H = H_0 \oplus \bigoplus_{r=1}^m (E_r \otimes K_r),$$

where each E_r is finite dimensional, F acts as 0 on H_0 , and on $E_r \otimes K_r$ it acts as $\rho_r(F) \otimes 1_{K_r}$ for a finite-dimensional representation ρ_r . If $H_0 \neq 0$, include it as the extra index $r = 0$ by putting $E_0 = \mathbb{C}$, $K_0 = H_0$, and $\rho_0 = 0$. Since there are only finitely many summands and H is nonseparable, choose an index r for which K_r is nonseparable. Write

$$E = E_r, \quad K = K_r, \quad \rho = \rho_r.$$

By Lemma 1, the closed subspace

$$M = \overline{\text{span}} \bigcup_{n=1}^{\infty} (T_n(H) \cup T_n^*(H))$$

is separable. Let P_r be the projection onto the chosen summand $E \otimes K$. Since E is finite dimensional, there is a separable subspace $N \subseteq K$ such that

$$P_r M \subseteq E \otimes N.$$

Choose a unit vector $\eta \in K \ominus N$, and put

$$V = E \otimes \mathbb{C}\eta.$$

Then $V \perp M$. Hence $T_n V = T_n^* V = 0$, so V reduces every T_n . It also reduces F , and the F -block on V is $\rho(f_n)$. Choose any rank-one projection p whose range is contained in V . Then $p \in \mathcal{K}(H) \subseteq A_F$ and $\|p\| = 1$.

Let $b_n = a_n|_{V^\perp}$. Because a_n is block diagonal with respect to $H = V \oplus V^\perp$,

$$\|a_n\| = \max(\|\rho(f_n)\|, \|b_n\|).$$

The space $K' = K \ominus (N + \mathbb{C}\eta)$ is infinite dimensional. On $E \otimes K' \subseteq V^\perp$, the operator b_n agrees with

$$\rho(f_n) \otimes 1_{K'} + T_n|_{E \otimes K'}.$$

The second term is compact, so Lemma 2 gives $\|b_n\| \geq \|\rho(f_n)\|$. Consequently $\|a_n\| = \|b_n\|$.

Finally $p - a_n$ is also block diagonal for $H = V \oplus V^\perp$, with blocks $p|_V - \rho(f_n)$ and $-b_n$. Thus

$$\|p - a_n\| = \max(\|p|_V - \rho(f_n)\|, \|b_n\|) \geq \|b_n\| = \|a_n\|.$$

The unit-sphere element p is not in any open ball $B(a_n, \|a_n\|)$. Since the countable family of centers was arbitrary, A_F fails BCP. $\square$

Scope and Limitations

This theorem does not solve the arbitrary C^* -algebra classification, but it supplies a sharp type-I obstruction. The previous continuous-trace theorem required separable compact-operator fibres. The present result proves that dropping fibre separability is impossible: even the elementary algebra $\mathcal{K}(H)$ has one primitive ideal when H is nonseparable, yet fails BCP. The finite-dimensional extension clause strengthens the obstruction: adding any finite-dimensional algebra $F \subseteq \mathcal{B}(H)$ does not repair BCP.

This also gives a simpler type-I analogue of the earlier one-point primitive obstruction from II_1 factors. The obstruction here is not diffuseness; it is the availability of a new rank-one projection orthogonal to the separable support of any countable list of compact centers.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, compact operators, unitization, finite-dimensional extension, nonseparable Hilbert space, $\mathcal{K}(H)$, one-point primitive space, finite primitive space, BCP, and UBCP. The existing packets included the continuous-trace separable-fibre positive theorem and the II_1 -factor one-point primitive obstruction, but no compact-operator fibre-separability obstruction. Web searches on June 17, 2026 for BCP and compact operators found the source paper and adjacent operator-space BCP literature, but no exact statement of Theorem 1.

Human Review Focus

The proof is low-tech. Reviewers should check Lemma 1 for compact operators on nonseparable Hilbert spaces, Lemma 2 for compact perturbations of a finite-dimensional matrix block, the finite-dimensional representation decomposition of F , and the final block-diagonal norm computation. Once those are accepted, the BCP failure is immediate.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Run `fa_banach_001`, lane 19, *The Separable C^* -Algebra Case of BCP*, solution packet `2311.14257_separable_cstar_bcp_classification`, 2026.
- [3] Run `fa_banach_001`, lane 19, *BCP for Continuous-Trace C^* -Algebras with Separable Fibres*, solution packet `2311.14257_continuous_trace_separable_fiber_bcp`, 2026.